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United States Patent Application Publication

20250283844

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

WATANABE; Yusuke et al.

GAS SENSOR, AND CONCENTRATION MEASUREMENT METHOD USING GAS SENSOR

Abstract

A sensor element includes first to fourth chambers communicating sequentially, an adjustment cell pumps oxygen out of a measurement gas introduced into the first chamber so that NO_x, H₂O, and CO₂ are not decomposed, a first measurement cell pumps out oxygen from the second chamber so that all NO_x is reduced, a second measurement cell pumps out oxygen from the third chamber so that all H₂O and CO₂ are reduced, a third measurement cell pumps oxygen into the fourth chamber to selectively oxidize H₂ generated by reduction, a concentration of NO_x is identified from a current generated by pumping-out by the first measurement pump cell, a concentration of H₂O is identified from a current generated by pumping-in by the third measurement pump cell, and a concentration of CO₂ is identified based on the identified concentration of H₂O and a current generated by pumping-out by the second measurement pump cell.

Inventors: WATANABE; Yusuke (Nagoya-shi, JP), ICHIKAWA; Daichi (Nagoya-shi, JP)

Applicant: NGK INSULATORS, LTD. (Nagoya-shi, JP)

Family ID: 91195387

Appl. No.: 19/214316

Filed: May 21, 2025

Foreign Application Priority Data

JP	2022-187223	Nov. 24, 2022
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Related U.S. Application Data

parent WO continuation PCT/JP2023/033742 20230915 PENDING child US 19214316

Publication Classification

Int. Cl.: G01N27/27 (20060101); G01N27/407 (20060101); G01N27/41 (20060101);
G01N27/417 (20060101); G01N33/00 (20060101)

U.S. Cl.:

CPC G01N27/27 (20130101); G01N27/4075 (20130101); G01N27/41 (20130101);
G01N27/417 (20130101); G01N33/0037 (20130101); G01N33/004 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of PCT/JP2023/033742, filed on Sep. 15, 2023, which claims the benefit of priority of Japanese Application No. JP2022-187223, filed on Nov. 24, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present invention relates to a multi-gas sensor capable of sensing a plurality of types of sensing target gas components and measuring concentrations thereof.

Description of the Background Art

[0003] In measurement for managing the amount of an emitted exhaust gas from a vehicle, technology of measuring concentrations of water vapor (H.sub.2O) and carbon dioxide (CO.sub.2) has already been known (see Japanese Patent No. 5918177 and No. 6469464, for example). In each of gas sensors disclosed in Japanese Patent No. 5918177 and No. 6469464, a water vapor (H.sub.2O) component and a carbon dioxide (CO.sub.2) component can be measured in parallel.

[0004] A gas sensor (NO_x sensor) capable of measuring NO_x by performing different pump cell control from that performed in the gas sensors disclosed in Japanese Patent No. 5918177 and No. 6469464 while including a sensor element having a similar configuration to that of the gas sensors disclosed in Japanese Patent No. 5918177 and No. 6469464 has also already been known (see Japanese Patent No. 3798412 for example).

[0005] In the gas sensor disclosed in Japanese Patent No. 5918177 having a three-chamber configuration, firstly, a main pump cell as a pump cell for a first internal space operates to pump out O₂ contained in a measurement gas introduced into the first internal space and to reduce all H₂O and CO₂ similarly contained in the measurement gas once to generate H₂ and CO. The measurement gas containing these H₂ and CO is introduced into a second internal space and further into a third internal space. A first measurement pump cell as a pump cell for the second internal space then pumps in O₂ to selectively oxidize H₂ to generate H₂O, and, further, a second measurement pump cell as a pump cell for the third internal space pumps in O₂ to oxidize CO to generate CO₂. Concentrations of H₂O and CO₂ in the measurement gas are respectively measured based on magnitudes of pump currents flowing through the first measurement pump cell and the second measurement pump cell when H₂ and CO are oxidized.

[0006] In the gas sensor, an applied voltage in the pump cell for the first internal space is required to be set to be high for reduction of H₂O and CO₂ in the first internal space. In addition, a temperature of a main inner pump electrode as an in-space pump electrode forming the main pump cell is required to be set to be high. Such a high applied voltage and maintaining the pump electrode at a high temperature, however, might cause a sensor element containing oxygen-ion

conductive solid electrolyte ceramics as a major component to be cracked and blackened to reduce the solid electrolyte ceramics.

[0007] Furthermore, measurement of NO_x is sometimes desired to be performed in parallel with measurement of H₂O and CO contained in an exhaust gas. In particular, in terms of ease of securement of an attachment space and cost efficiency, there is a need to simultaneously measure H₂O and CO as well as NO_x using the same gas sensor.

SUMMARY

[0008] The present invention relates to a multi-gas sensor capable of sensing a plurality of types of sensing target gas components and measuring concentrations thereof.

[0009] According to the present invention, a gas sensor capable of measuring concentrations of a plurality of sensing target gas components, includes: a sensor element having a structure formed of an oxygen-ion conductive solid electrolyte; and a controller controlling operation of the gas sensor, wherein the sensor element includes: a gas inlet through which a measurement gas is introduced; a first chamber, a second chamber, a third chamber, and a fourth chamber communicating sequentially from the gas inlet via different diffusion control parts; an adjustment pump cell including an inner electrode formed to face the first chamber, an out-of-space pump electrode provided at a location other than a location in the first chamber, the second chamber, the third chamber, and the fourth chamber, and a portion of the solid electrolyte present between the inner electrode and the out-of-space pump electrode; a first measurement pump cell including a first measurement electrode formed to face the second chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the first measurement electrode and the out-of-space pump electrode; a second measurement pump cell including a second measurement electrode formed to face the third chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the second measurement electrode and the out-of-space pump electrode; a third measurement pump cell including a third measurement electrode formed to face the fourth chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the third measurement electrode and the out-of-space pump electrode; and a heater heating the sensor element, the inner electrode is a cermet electrode containing a Pt—Au alloy as a metal component, the Pt—Au alloy having an Au concentration of 0.5 wt % or more, the first measurement electrode is another cermet electrode containing a Pt—Rh alloy as a metal component, the adjustment pump cell pumps oxygen out of the measurement gas introduced through the gas inlet into the first chamber so that NO_x, water vapor, and carbon dioxide contained in the measurement gas are not decomposed, the first measurement pump cell pumps out oxygen from the second chamber so that substantially all NO_x contained in the measurement gas introduced from the first chamber into the second chamber is reduced, the second measurement pump cell pumps out oxygen from the third chamber so that substantially all water vapor and carbon dioxide contained in the measurement gas introduced from the second chamber into the third chamber are reduced, the third measurement pump cell pumps oxygen into the fourth chamber to selectively oxidize, in the fourth chamber, hydrogen having been generated by reduction of water vapor and contained in the measurement gas introduced from the third chamber into the fourth chamber, and the controller identifies: a concentration of NO_x contained in the measurement gas based on a magnitude of a NO_x detection current as an oxygen pump current flowing between the first measurement electrode and the out-of-space pump electrode when oxygen is pumped out from the second chamber by the first measurement pump cell; a concentration of water vapor contained in the measurement gas based on a value of a water vapor equivalent current as an oxygen pump current flowing between the third measurement electrode and the out-of-space pump electrode when hydrogen is oxidized with oxygen pumped into the forth chamber by the third measurement pump cell; and a concentration of carbon dioxide contained in the measurement gas based on the value of the water vapor equivalent current and a value of a total reducing current as an oxygen pump current flowing between the second measurement electrode and the out-of-space

pump electrode when water vapor and carbon dioxide are reduced by the second measurement pump cell pumping out oxygen from the third chamber.

[0010] Another aspect of the present invention is a concentration measurement method of measuring concentrations of a plurality of sensing target gas components using a gas sensor, wherein the gas sensor includes a sensor element having an elongated planar structure formed of an oxygen-ion conductive solid electrolyte, the sensor element includes: a gas inlet through which a measurement gas is introduced; a first chamber, a second chamber, a third chamber, and a fourth chamber communicating sequentially from the gas inlet via different diffusion control parts; an adjustment pump cell including an inner electrode formed to face the first chamber, an out-of-space pump electrode provided at a location other than a location in the first chamber, the second chamber, the third chamber, and the fourth chamber, and a portion of the solid electrolyte present between the inner electrode and the out-of-space pump electrode; a first measurement pump cell including a first measurement electrode formed to face the second chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the first measurement electrode and the out-of-space pump electrode; a second measurement pump cell including a second measurement electrode formed to face the third chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the second measurement electrode and the out-of-space pump electrode; a third measurement pump cell including a third measurement electrode formed to face the fourth chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the third measurement electrode and the out-of-space pump electrode; and a heater heating the sensor element, the inner electrode is a cermet electrode containing a Pt—Au alloy as a metal component, the Pt—Au alloy having an Au concentration of 0.5 wt % or more, the first measurement electrode is another cermet electrode containing a Pt—Rh alloy as a metal component, and the concentration measurement method using the gas sensor includes: a) pumping, using the adjustment pump cell, oxygen out of the measurement gas introduced through the gas inlet into the first chamber so that NO_x, water vapor, and carbon dioxide contained in the measurement gas are not decomposed; b) pumping out, using the first measurement pump cell, oxygen from the second chamber so that substantially all NO_x contained in the measurement gas introduced from the first chamber into the second chamber is reduced; c) pumping out, using the second measurement pump cell, oxygen from the third chamber so that substantially all water vapor and carbon dioxide contained in the measurement gas introduced from the second chamber into the third chamber are reduced; d) pumping, using the third measurement pump cell, oxygen into the fourth chamber to selectively oxidize, in the fourth chamber, hydrogen having been generated by reduction of water vapor and contained in the measurement gas introduced from the third chamber into the fourth chamber; e) identifying a concentration of NO_x contained in the measurement gas based on a magnitude of a NO_x detection current as an oxygen pump current flowing between the first measurement electrode and the out-of-space pump electrode when NO_x is reduced by pumping out oxygen from the second chamber using the first measurement pump cell; f) identifying a concentration of water vapor contained in the measurement gas based on a value of a water vapor equivalent current as an oxygen pump current flowing between the third measurement electrode and the out-of-space pump electrode when hydrogen is oxidized with oxygen pumped into the fourth chamber using the third measurement pump cell; and g) identifying a concentration of carbon dioxide contained in the measurement gas based on the value of the water vapor equivalent current and a value of a total reducing current as an oxygen pump current flowing between the second measurement electrode and the out-of-space pump electrode when water vapor and carbon dioxide are reduced by pumping out oxygen from the third chamber using the second measurement pump cell.

[0011] According to the present invention, a multi-gas sensor having more long-term reliability than before and capable of simultaneously measuring many types of gases is implemented.

[0012] It is therefore an object of the present invention to provide a multi-gas sensor with more

long-term reliability than before capable of simultaneously measuring a water vapor (H.sub.2O) component and a carbon dioxide (CO.sub.2) component as well as NOx, suppressing cracking and blackening of a sensor element, and further being less likely to be subjected to a change in sensitivity during a long-term use.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a diagram schematically showing one example of a configuration of a gas sensor **100**.

[0014] FIG. 2 is a block diagram showing functional components implemented by a controller **110**.

[0015] FIG. 3 is a schematic diagram illustrating entry and exit of gases into and from four chambers comprised in a sensor element **101** of the gas sensor **100**.

[0016] FIG. 4 is a graph showing a relationship between a target value of electromotive force **V0** in a first chamber sensor cell **80** and an oxygen pump current **Ip0** flowing through an adjustment pump cell **21** when three different types of model gases are caused to flow.

[0017] FIG. 5 is a diagram showing dependence of an oxygen pump current **Ip2** on a concentration of a sensing target gas component.

[0018] FIG. 6 is a diagram showing dependence of an oxygen pump current **Ip3** on a concentration of a sensing target gas component.

[0019] FIG. 7 is a diagram showing an example of H.sub.2O characteristics data.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Configuration of Gas Sensor

[0020] FIG. 1 is a diagram schematically showing one example of a configuration of a gas sensor **100** according to the present embodiment. The gas sensor **100** is a multi-gas sensor sensing a plurality of types of gas components and measuring concentrations thereof using a sensor element **101**. Assume that at least water vapor (H.sub.2O), carbon dioxide (CO.sub.2), and nitrogen oxide (NOx) are main sensing target gas components of the gas sensor **100** in the present embodiment. The gas sensor **100** is attached to an exhaust path of an internal combustion engine, such as an engine of a vehicle, and is used with an exhaust gas flowing along the exhaust path as a measurement gas, for example. FIG. 1 includes a vertical cross-sectional view taken along a longitudinal direction of the sensor element **101**.

[0021] The sensor element **101** includes an elongated planar structure (base part) **14** formed of an oxygen-ion conductive solid electrolyte, a first diffusion control part **11** doubling a gas inlet **10** which is formed in one end portion (a left end portion in the figure) of the structure **14** and through which the measurement gas is introduced, and a buffer space **12**, a first chamber **20**, a second chamber **40**, a third chamber **61**, and a fourth chamber **63** formed in the structure **14** and communicating sequentially from the gas inlet **10** (first diffusion control part **11**). The buffer space **12** communicates with the gas inlet **10** (first diffusion control part **11**). The first chamber **20** communicates with the buffer space **12** via a second diffusion control part **13**. The second chamber **40** communicates with the first chamber **20** via a third diffusion control part **30**. The third chamber **61** communicates with the second chamber **40** via a fourth diffusion control part **60**. The fourth chamber **63** communicates with the third chamber **61** via a fifth diffusion control part **62**.

[0022] The structure **14** is formed by laminating a plurality of substrates of ceramics, for example. Specifically, the structure **14** has a configuration in which six layers including a first substrate **1**, a second substrate **2**, a third substrate **3**, a first solid electrolyte layer **4**, a spacer layer **5**, and a second solid electrolyte layer **6** are sequentially laminated from the bottom. Each layer is formed of an oxygen-ion conductive solid electrolyte, such as zirconia (ZrO.sub.2).

[0023] The first diffusion control part **11** doubling as the gas inlet **10**, the buffer space **12**, the

second diffusion control part **13**, the first chamber **20**, the third diffusion control part **30**, the second chamber **40**, the fourth diffusion control part **60**, the third chamber **61**, the fifth diffusion control part **62**, and the fourth chamber **63** are formed in this order between a lower surface **6b** of the second solid electrolyte layer **6** and an upper surface **4a** of the first solid electrolyte layer **4** on a side of the one end portion of the structure **14**. A part extending from the gas inlet **10** to the fourth chamber **63** is also referred to as a gas distribution part.

[0024] The buffer space **12**, the first chamber **20**, the second chamber **40**, the third chamber **61**, and the fourth chamber **63** are formed to penetrate the spacer layer **5** in a thickness direction. The lower surface **6b** of the second solid electrolyte layer **6** is exposed in upper portions in the figure of these chambers and the like, and the upper surface **4a** of the first solid electrolyte layer **4** is exposed in lower portions in the figure of these chambers and the like. Side portions of these chambers and the like are each defined by the spacer layer **5** or any of the diffusion control parts. The first chamber **20**, the second chamber **40**, the third chamber **61**, and the fourth chamber **63** each have a length (size in the longitudinal direction of the element) of 0.3 mm to 1.0 mm, for example, a width (size in a transverse direction of the element) of 0.5 mm to 30 mm, for example, and a height (size in a thickness direction of the element) of 50 μm to 200 μm , for example. These chambers, however, are not required to have the same size and may have different sizes.

[0025] The gas inlet **10** may similarly be formed to penetrate the spacer layer **5** in the thickness direction separately from the first diffusion control part **11**. In this case, the first diffusion control part **11** is to be formed inside and adjacent to the gas inlet **10**.

[0026] The first diffusion control part **11**, the second diffusion control part **13**, the third diffusion control part **30**, the fourth diffusion control part **60**, and the fifth diffusion control part **62** each include two horizontally long slits. That is to say, they each have openings elongated in a direction perpendicular to the page of the figure in an upper portion and a lower portion in the figure thereof. The slits each have a length (size in the longitudinal direction of the element) of 0.2 mm to 1.0 mm, for example, a width of an opening (size in the transverse direction of the element) of 0.5 mm to 30 mm, for example, and a height of the opening (size in the thickness direction of the element) of 5 μm to 30 μm , for example.

[0027] The sensor element **101** includes a reference gas introduction space **43** in the other end portion (a right end portion in the figure) opposite the one end portion in which the gas inlet **10** is provided. The reference gas introduction space **43** is formed between an upper surface **3a** of the third substrate **3** and a lower surface **5b** of the spacer layer **5**. A side portion of the reference gas introduction space **43** is defined by a side surface of the first solid electrolyte layer **4**. Oxygen (O_{2}) and air are introduced into the reference gas introduction space **43** as reference gases, for example.

[0028] The gas inlet **10** (first diffusion control part **11**) is a part opening to an external space, and the measurement gas is taken from the external space into the sensor element **101** through the gas inlet **10**.

[0029] The first diffusion control part **11** is a part providing predetermined diffusion resistance to the taken measurement gas.

[0030] The buffer space **12** is provided to cancel concentration fluctuations of the measurement gas caused by pressure fluctuations of the measurement gas in the external space. Pulsation of exhaust pressure of the exhaust gas of the vehicle is taken as an example of such pressure fluctuations of the measurement gas, for example.

[0031] The second diffusion control part **13** is a part providing predetermined diffusion resistance to the measurement gas introduced from the buffer space **12** into the first chamber **20**.

[0032] The first chamber **20** is provided as a space to pump oxygen out of the measurement gas introduced through the second diffusion control part **13**. Pumping-out of oxygen is implemented by operation of an adjustment pump cell **21**.

[0033] The adjustment pump cell **21** is an electrochemical pump cell including an inner pump

electrode (adjustment electrode) **22**, an outer pump electrode (out-of-space pump electrode) **23**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes. [0034] In the adjustment pump cell **21**, a voltage V_{p0} is applied across the inner pump electrode **22** and the outer pump electrode **23** from a variable power supply **24** disposed outside the sensor element **101** to generate an oxygen pump current (oxygen ion current) I_{p0} . Oxygen in the first chamber **20** can thereby be pumped out to the external space. Assume that a direction of the oxygen pump current I_{p0} when oxygen is pumped out from the first chamber **20** is a positive direction of the oxygen pump current I_{p0} in the present embodiment.

[0035] The inner pump electrode **22** is provided on substantially the entire portions of the lower surface **6b** of the second solid electrolyte layer **6** and the upper surface **4a** of the first solid electrolyte layer **4** defining the first chamber **20** respectively as a ceiling electrode portion **22a** and a bottom electrode portion **22b**. The ceiling electrode portion **22a** and the bottom electrode portion **22b** are connected by an unillustrated conducting portion.

[0036] The inner pump electrode **22** is provided, with an alloy (a Pt—Au alloy) of platinum (Pt) and gold (Au) inert to NO_x as a metal component, as a porous cermet electrode containing the Pt—Au alloy and zirconia and being rectangular in plan view, for example. The Pt—Au alloy preferably contains Au at a concentration of 0.5 wt % or more to surely pump out only oxygen contained in the measurement gas without reducing NO_x , H_2O , and CO_2 .

[0037] The outer pump electrode **23** is provided, with platinum or the Pt—Au alloy as a metal component, as a porous cermet electrode containing platinum or the Pt—Au alloy and zirconia and being rectangular in plan view, for example.

[0038] In the sensor element **101**, the inner pump electrode **22**, a reference electrode **42**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes constitute a first chamber sensor cell **80**. The first chamber sensor cell **80** is an electrochemical sensor cell to grasp oxygen partial pressure in an atmosphere in the first chamber **20**.

[0039] The reference electrode **42** is an electrode formed between the first solid electrolyte layer **4** and the third substrate **3** and is provided as a porous cermet electrode containing platinum and zirconia and being rectangular in plan view, for example.

[0040] A reference gas introduction layer **48** formed of porous alumina and leading to the reference gas introduction space **43** is provided around the reference electrode **42**. A reference gas in the reference gas introduction space **43** is introduced into a surface of the reference electrode **42** via the reference gas introduction layer **48**. That is to say, the reference electrode **42** is always in contact with the reference gas.

[0041] In the first chamber sensor cell **80**, electromotive force (Nernst electromotive force) V_0 is generated between the inner pump electrode **22** and the reference electrode **42**. The electromotive force V_0 has a value in accordance with a difference between an oxygen concentration (oxygen partial pressure) in the first chamber **20** and an oxygen concentration (oxygen partial pressure) of the reference gas. The oxygen concentration (oxygen partial pressure) of the reference gas is basically constant, so that the electromotive force V_0 has a value in accordance with the oxygen concentration (oxygen partial pressure) in the first chamber **20**.

[0042] The third diffusion control part **30** is a part providing predetermined diffusion resistance to the measurement gas introduced from the first chamber **20** into the second chamber **40**.

[0043] The second chamber **40** is provided as a space to reduce (decompose) NO_x contained as a sensing target gas component in the measurement gas introduced through the third diffusion control part **30** and pump out oxygen thus generated, so that the measurement gas contains H_2O and CO_2 but does not substantially contain NO_x . Pumping-out of oxygen is implemented by operation of a first measurement pump cell **50**.

[0044] The first measurement pump cell **50** is an electrochemical pump cell including a first measurement electrode **51**, the outer pump electrode **23**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes.

[0045] In the first measurement pump cell **50**, a voltage V_{p1} is applied across the first measurement electrode **51** and the outer pump electrode **23** from a variable power supply **52** disposed outside the sensor element **101** to generate an oxygen pump current (oxygen ion current) I_{p1} . Oxygen in the second chamber **40** can thereby be pumped out to the external space. Assume that a direction of the oxygen pump current I_{p1} when oxygen is pumped out from the second chamber **40** is a positive direction of the oxygen pump current I_{p1} in the present embodiment.

[0046] The first measurement electrode **51** is provided on substantially the entire portions of the lower surface **6b** of the second solid electrolyte layer **6** and the upper surface **4a** of the first solid electrolyte layer **4** defining the second chamber **40** respectively as a ceiling electrode portion **51a** and a bottom electrode portion **51b**. The ceiling electrode portion **51a** and the bottom electrode portion **51b** are connected by an unillustrated conducting portion.

[0047] The first measurement electrode **51** is provided, with an alloy (a Pt—Rh alloy) of platinum and rhodium (Rh) as a metal component, as a porous cermet electrode containing the Pt—Rh alloy and zirconia and being rectangular in plan view, for example. The Pt—Rh alloy preferably has an Rh concentration of 30 wt % or more.

[0048] In the sensor element **101**, the first measurement electrode **51**, the reference electrode **42**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes constitute a second chamber sensor cell **81**. The second chamber sensor cell **81** is an electrochemical sensor cell to grasp oxygen partial pressure in an atmosphere in the second chamber **40**.

[0049] In the second chamber sensor cell **81**, electromotive force (Nernst electromotive force) V_1 is generated between the first measurement electrode **51** and the reference electrode **42**. The electromotive force V_1 has a value in accordance with a difference between an oxygen concentration (oxygen partial pressure) in the second chamber **40** and the oxygen concentration (oxygen partial pressure) of the reference gas. Since the oxygen concentration (oxygen partial pressure) of the reference gas is basically constant, the electromotive force V_1 has a value in accordance with the oxygen concentration (oxygen partial pressure) in the second chamber **40**.

[0050] The fourth diffusion control part **60** is a part providing predetermined diffusion resistance to the measurement gas introduced from the second chamber **40** into the third chamber **61**, containing H_2O and CO_2 , and not substantially containing NO_x and oxygen.

[0051] The third chamber **61** is provided as a space to reduce (decompose) H_2O and CO_2 contained as sensing target gas components in the measurement gas introduced through the fourth diffusion control part **60** to generate hydrogen (H_2) and carbon monoxide (CO), so that the measurement gas does not substantially contain NO_x and oxygen as well as H_2O and CO_2 . Reduction (decomposition) of H_2O and CO_2 is implemented by operation of a second measurement pump cell **41**.

[0052] The second measurement pump cell **41** is an electrochemical pump cell including a second measurement electrode **44**, the outer pump electrode **23**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes.

[0053] In the second measurement pump cell **41**, a voltage V_{p2} is applied across the second measurement electrode **44** and the outer pump electrode **23** from a variable power supply **46** disposed outside the sensor element **101** to generate an oxygen pump current (oxygen ion current) I_{p2} . Oxygen generated in the third chamber **61** by reduction of H_2O and CO_2 can thereby be pumped out to the external space. Assume that a direction of the oxygen pump current I_{p2} when oxygen is pumped out from the third chamber **61** is a positive direction of the oxygen pump current I_{p2} in the present embodiment.

[0054] The second measurement electrode **44** is provided on substantially the entire portions of the lower surface **6b** of the second solid electrolyte layer **6** and the upper surface **4a** of the first solid electrolyte layer **4** defining the third chamber **61** respectively as a ceiling electrode portion **44a** and a bottom electrode portion **44b**. The ceiling electrode portion **44a** and the bottom electrode portion

44b are connected by an unillustrated conducting portion.

[0055] The second measurement electrode **44** is provided as a porous cermet electrode containing Pt as a metal component and being rectangular in plan view.

[0056] In the sensor element **101**, the second measurement electrode **44**, the reference electrode **42**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes constitute a third chamber sensor cell **82**. The third chamber sensor cell **82** is an electrochemical sensor cell to grasp oxygen partial pressure in an atmosphere in the third chamber **61**.

[0057] In the third chamber sensor cell **82**, electromotive force (Nernst electromotive force) **V2** is generated between the second measurement electrode **44** and the reference electrode **42**. The electromotive force **V2** has a value in accordance with a difference between an oxygen concentration (oxygen partial pressure) in the third chamber **61** and the oxygen concentration (oxygen partial pressure) of the reference gas. Since the oxygen concentration (oxygen partial pressure) of the reference gas is basically constant, the electromotive force **V2** has a value in accordance with the oxygen concentration (oxygen partial pressure) in the third chamber **61**.

[0058] The fifth diffusion control part **62** is a part providing predetermined diffusion resistance to the measurement gas introduced from the third chamber **61** into the fourth chamber **63** and containing H.sub.2 and CO while not substantially containing H.sub.2O, CO.sub.2, NO_x, and oxygen.

[0059] The fourth chamber **63** is provided as a space to selectively oxidize all of H.sub.2 from among H.sub.2 and CO contained in the measurement gas introduced through the fifth diffusion control part **62** to generate H.sub.2O again. Generation of H.sub.2O by oxidation of H.sub.2 is implemented by operation of a third measurement pump cell **66**.

[0060] The third measurement pump cell **66** is an electrochemical pump cell including a third measurement electrode **64**, the outer pump electrode **23**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes.

[0061] In the third measurement pump cell **66**, a voltage **Vp3** is applied across the third measurement electrode **64** and the outer pump electrode **23** from a variable power supply **68** disposed outside the sensor element **101** to generate an oxygen pump current (oxygen ion current) **Ip3**. Oxygen can thereby be pumped into the fourth chamber **63** from the external space. Assume that a direction of the oxygen pump current **Ip3** when oxygen is pumped out from the fourth chamber **63** is a positive direction of the oxygen pump current **Ip3** in the present embodiment.

[0062] The third measurement electrode **64** is provided on substantially the entire portion of the upper surface **4a** of the first solid electrolyte layer **4** defining the fourth chamber **63**.

[0063] The third measurement electrode **64** is provided, with the Pt—Au alloy as a metal component, as a porous cermet electrode containing the Pt—Au alloy and zirconia and being rectangular in plan view. The Pt—Au alloy preferably has an Au concentration of 1 wt % or more and 50 wt % or less and more preferably has an Au concentration of 10 wt % or more and 30 wt % or less. In this case, a selective H.sub.2 oxidation property, that is, a property that, when H.sub.2 and CO coexist in the fourth chamber **63**, only H.sub.2 is selectively oxidized with oxygen pumped in by the third measurement pump cell **66** and CO is not oxidized, of the third measurement electrode **64** is more suitably developed.

[0064] In the sensor element **101**, the third measurement electrode **64**, the reference electrode **42**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes constitute a fourth chamber sensor cell **83**. The fourth chamber sensor cell **83** is an electrochemical sensor cell to grasp oxygen partial pressure in an atmosphere in the fourth chamber **63**.

[0065] In the fourth chamber sensor cell **83**, electromotive force (Nernst electromotive force) **V3** is generated between the third measurement electrode **64** and the reference electrode **42**. The electromotive force **V3** has a value in accordance with a difference between an oxygen concentration (oxygen partial pressure) in the fourth chamber **63** and the oxygen concentration (oxygen partial pressure) of the reference gas. Since the oxygen concentration (oxygen partial

pressure) of the reference gas is basically constant, the electromotive force V_3 has a value in accordance with the oxygen concentration (oxygen partial pressure) in the fourth chamber **63**.
[0066] The sensor element **101** further includes an electrochemical sensor cell **84** including the outer pump electrode **23**, the reference electrode **42**, and a solid electrolyte present in a portion of the structure **14** sandwiched between these electrodes. Electromotive force V_{ref} generated between the outer pump electrode **23** and the reference electrode **42** of the sensor cell **84** has a value in accordance with oxygen partial pressure of the measurement gas present outside the sensor element **101**.

[0067] In addition to the foregoing, the sensor element **101** includes a heater part **70** playing a role in temperature adjustment of heating the sensor element **101** and maintaining the temperature thereof to enhance oxygen ion conductivity of the solid electrolyte forming the structure **14**.

[0068] The heater part **70** mainly includes a heater electrode **71**, a heater element **72**, a heater lead **72a**, a through hole **73**, a heater insulating layer **74**, and a heater resistance detection lead, which is not illustrated in FIG. **1**. The heater element **72** is hereinafter also simply referred to as a heater **72**.

[0069] The heater **72** is provided to be sandwiched between the second substrate **2** and the third substrate **3** from below and above and generates heat by being powered from outside through the heater electrode **71** provided on a lower surface **1b** of the first substrate **1**, the through hole **73**, and the heater lead **72a**. The heater **72** is buried over the entire region of a range from the buffer space **12** to the fourth chamber **63** and can heat the sensor element **101** to a predetermined temperature and, further, maintain the temperature.

[0070] The heater **72** is provided so that a temperature is highest near the first chamber **20** (near the inner pump electrode **22**) and decreases with increasing distance from the first chamber **20** in the longitudinal direction of the element during heating. In the present embodiment, a temperature in a range from the one end portion of the sensor element **101** in which the gas inlet **10** is disposed to the fourth chamber **63** when the gas sensor **100** is in use (when the sensor element **101** is driven) is referred to as an element driving temperature. The heater **72** performs heating so that the element driving temperature is in a range of 750° C. to 950° C.

[0071] The heater insulating layer **74** of alumina and the like is formed above and below the heater **72** to electrically insulate the heater **72** from the second substrate **2** and the third substrate **3**. The heater part **70** also includes a pressure dissipation hole **75**. The pressure dissipation hole **75** is a part provided to penetrate the third substrate **3** and communicate with the reference gas introduction space **43** and is provided to mitigate a rise in internal pressure associated with a rise in temperature in the heater insulating layer **74**.

[0072] The gas sensor **100** further includes a controller **110** controlling operation of the sensor element **101** and performing processing to identify concentrations of the sensing target gas components based on currents flowing through the sensor element **101**.

[0073] FIG. **2** is a block diagram showing functional components implemented by the controller **110**. The controller **110** is configured by one or more electronic circuits including one or more central processing units (CPUs), a storage device, and the like, for example. Each of the electronic circuits is a software functional part implementing a predetermined functional component by a CPU executing a predetermined program stored in the storage device, for example. The controller **110** may naturally be configured by an integrated circuit, such as a field-programmable gate array (FPGA), on which a plurality of electronic circuits are connected in accordance with their functions and the like.

[0074] When the gas sensor **100** is attached to the exhaust path of the engine of the vehicle and is used with the exhaust gas flowing along the exhaust path as the measurement gas, some or all of functions of the controller **110** may be implemented by an electronic control unit (ECU) of the vehicle.

[0075] The controller **110** includes, as functional components implemented by the CPU executing a predetermined program, an element operation control part **120** controlling operation of each part of

the sensor element **101** described above and a concentration identification part **130** performing processing to identify the concentrations of the sensing target gas components contained in the measurement gas.

[0076] The element operation control part **120** mainly includes an adjustment pump cell control part **121** controlling operation of the adjustment pump cell **21**, a first measurement pump cell control part **122a** controlling operation of the first measurement pump cell **50**, a second measurement pump cell control part **122b** controlling operation of the second measurement pump cell **41**, a third measurement pump cell control part **122c** controlling operation of the third measurement pump cell **66**, and a heater control part **123** controlling heating operation performed by the heater **72**.

[0077] On the other hand, the concentration identification part **130** mainly includes a NO_x concentration identification part **130N**, a water vapor concentration identification part **130H**, and a carbon dioxide concentration identification part **130C** respectively identifying concentrations of NO_x, H₂O, and CO₂ as the main sensing target gas components of the gas sensor **100**.

[0078] The NO_x concentration identification part **130N** identifies the concentration of NO_x contained in the measurement gas based on a value of the oxygen pump current **Ip1** flowing through the first measurement pump cell **50** acquired by the first measurement pump cell control part **122a**.

[0079] The water vapor concentration identification part **130H** identifies the concentration of H₂O contained in the measurement gas based on a value of the oxygen pump current **Ip3** flowing through the third measurement pump cell **66** acquired by the third measurement pump cell control part **122c**.

[0080] The carbon dioxide concentration identification part **130C** identifies the concentration of CO₂ contained in the measurement gas based on the concentration of H₂O identified by the water vapor concentration identification part **130H** (the value of the oxygen pump current **Ip3** based on which the concentration of H₂O is identified) and a value of the oxygen pump current **Ip2** flowing through the second measurement pump cell **41** acquired by the second measurement pump cell control part **122b**.

[0081] The concentration identification part **130** further includes an oxygen concentration identification part **130A** identifying a concentration of oxygen contained in the measurement gas. The oxygen concentration identification part **130A** identifies the concentration of oxygen contained in the measurement gas based on a value of the oxygen pump current **Ip0** flowing through the adjustment pump cell **21** acquired by the adjustment pump cell control part **121**. That is to say, the gas sensor **100** according to the present embodiment senses, in addition to NO_x, H₂O, and CO₂ as the main sensing target gas components, oxygen as an appendant sensing target gas component.

Multi-Gas Sensing and Concentration Identification

[0082] A method of sensing a plurality of types of gases (multi-gas sensing) and identifying concentrations of the sensed gases implemented by the gas sensor **100** having a configuration as described above will be described next. Assume hereinafter that the measurement gas is an exhaust gas containing oxygen, NO_x, H₂O, and CO₂.

[0083] FIG. **3** is a schematic diagram illustrating entry and exit of gases into and from the four chambers (internal spaces) comprised in the sensor element **101** of the gas sensor **100**.

[0084] First, in the sensor element **101** of the gas sensor **100** according to the present embodiment, the measurement gas is introduced through the gas inlet **10** (first diffusion control part **11**), the buffer space **12**, and the second diffusion control part **13** into the first chamber **20** as described above. In the first chamber **20**, oxygen is pumped out of the introduced measurement gas by operation of the adjustment pump cell **21**.

[0085] Pumping-out of oxygen is performed in the way that the adjustment pump cell control part **121** of the controller **110** sets a target value (control voltage) of the electromotive force **V0** in the

first chamber sensor cell **80** to a value in a range of 300 mV to 500 mV (preferably 350 mV) and feedback-controls the voltage V_{p0} applied from the variable power supply **24** to the adjustment pump cell **21** in accordance with a difference between an actual value and the target value of the electromotive force V_0 so that the electromotive force V_0 is maintained at the target value. A value of the electromotive force V_0 significantly deviates from the target value when the measurement gas containing a large amount of oxygen reaches the first chamber **20**, for example, and thus the adjustment pump cell control part **121** controls the pump voltage V_{p0} applied from the variable power supply **24** to the adjustment pump cell **21** so that the deviation is reduced.

[0086] The adjustment pump cell **21** pumps out oxygen from the first chamber **20** in such a manner, so that oxygen partial pressure (concentration) in the first chamber **20** is maintained at a sufficiently low value to the extent that H_{2O} and CO_2 contained in the measurement gas are not reduced. For example, the oxygen concentration in the first chamber **20** is approximately 0.1 ppm to 0.00001 ppm by operation of the adjustment pump cell **21**.

[0087] FIG. **4** is a diagram for describing a reason why oxygen is pumped out to the extent that H_{2O} and CO_2 are not reduced by setting the target value of the electromotive force V_0 to the value in the range of 300 mV to 500 mV. Specifically, FIG. **4** is a graph showing a relationship between the target value (control voltage) of the electromotive force V_0 in the first chamber sensor cell **80** and the oxygen pump current I_{p0} flowing through the adjustment pump cell **21** when three different types of model gases are caused to flow. Specifically, the three types of model gases include a first gas containing oxygen of 10%, a second gas containing oxygen of 10% and CO_2 of 10%, and a third gas containing oxygen of 10% and H_{2O} of 10%. The balance of each of the gases is nitrogen (N_2). The element driving temperature is 800° C. or more, and a temperature of each of the model gases is 150° C.

[0088] It can be seen from FIG. **4** that, in a case of the first gas, the oxygen pump current I_{p0} is substantially constant when the control voltage is in a range of 0.3 V or more, while, in a case of the second gas and the third gas, a profile is substantially the same as that for the first gas when the control voltage is in a range of 0.7 V or less but the oxygen pump current I_{p0} increases again when the control voltage exceeds 0.7 V. The increase results from superimposition of reducing currents of H_{2O} or CO_2 flowing when H_{2O} or CO_2 contained in the measurement gas is reduced (decomposed) to generate oxygen.

[0089] As such, the target value of the electromotive force V_0 is set to the value in the range of 300 mV to 500 mV in the present embodiment.

[0090] Similarly to H_{2O} and CO_2 , NO_x is not reduced with pumping-out of oxygen by the adjustment pump cell **21**. This, however, is not due to setting of the target value of the electromotive force V_0 but because the inner pump electrode **22** contains Au inert to NO_x as described above.

[0091] As described above, in the gas sensor **100** according to the present embodiment, only pumping-out of oxygen in a manner that H_{2O} and CO_2 are not reduced is performed in the first chamber **20** to be at a highest temperature in the sensor element **101** during operation, and reduction of H_{2O} and CO_2 is not performed, in contrast to a gas sensor in conventional technology. Furthermore, NO_x is also not reduced.

[0092] The target value of the electromotive force V_0 in the first chamber sensor cell **80** set for pumping-out of oxygen is 300 mV to 500 mV, which is sufficiently smaller than a target value of 1000 mV to 1500 mV set when H_{2O} and CO_2 are reduced. An increase in pump voltage V_{p0} is thus suppressed compared with a voltage applied to a corresponding pump cell in the gas sensor in conventional technology accompanied by reduction of H_{2O} and CO_2 . Cracking and blackening attributable to application of a high voltage with the inner pump electrode **22** being maintained at a high temperature are thus suitably suppressed in the gas sensor **100** according to the present embodiment.

[0093] Since the inner pump electrode **22** contains Au inert to NO_x , NO_x is not reduced with

pumping-out of oxygen by the adjustment pump cell **21** even when the measurement gas contains NO_x.

[0094] The measurement gas of which only oxygen has been pumped out in the first chamber **20** to the extent that NO_x, H₂O, and CO₂ are not reduced is introduced into the second chamber **40**. NO_x contained in the measurement gas is reduced in the second chamber **40**. The first measurement electrode **51** disposed in the second chamber **40** and forming the first measurement pump cell **50** contains the Pt—Rh alloy as the metal component and does not contain Au inert to NO_x, so that reduction of NO_x progresses in the first measurement electrode **51**. That is to say, oxygen is further pumped out of the measurement gas of which oxygen has been pumped out in the first chamber **20** and which is then introduced into the second chamber **40** by operation of the first measurement pump cell **50**, so that a reduction (decomposition) reaction (e.g., $2\text{NO} \rightarrow \text{N}_2 + \text{O}_2$) of NO_x contained in the measurement gas progresses, and substantially all NO is decomposed into nitrogen and oxygen.

[0095] Reduction (decomposition) of NO_x and pumping-out of oxygen thus generated are performed in the way that the first measurement pump cell control part **122a** of the controller **110** sets a target value (control voltage) of the electromotive force **V1** in the second chamber sensor cell **81** to a value in a range of 350 mV to 700 mV (preferably 400 mV) and feedback-controls the voltage **Vp1** applied from the variable power supply **52** to the first measurement pump cell **50** in accordance with a difference between an actual value and the target value of the electromotive force **V1** so that the electromotive force **V1** is maintained at the target value.

[0096] The first measurement pump cell **50** operates in this manner, so that the oxygen partial pressure in the second chamber **40** is maintained at a similar value to or a slightly lower value than that in the first chamber **20**. It is approximately 10⁻⁸ atm when an equation **V2**=400 mV holds true, for example. The measurement gas thus no longer substantially contains NO_x and oxygen while containing H₂O and CO₂ (and further N₂).

[0097] In the gas sensor **100** according to the present embodiment, the concentration of NO_x in the measurement gas is identified based on the oxygen pump current **Ip1** flowing through the first measurement pump cell **50** during pumping-out of oxygen including reduction of NO_x.

[0098] The oxygen pump current **Ip1** (hereinafter also referred to as a NO_x detection current **Ip1**) in the first measurement pump cell **50** flows with pumping-out of oxygen generated by decomposition of NO_x contained in the measurement gas. A magnitude of the NO_x detection current **Ip1** is thus substantially proportional to the concentration of NO_x contained in the measurement gas introduced through the gas inlet **10**. That is to say, there is a linear relationship between the NO_x detection current **Ip1** and the concentration of NO_x in the measurement gas. Data (**Ip1**-NO_x data) indicating the linear relationship is identified in advance using model gases having known NO_x concentrations and is stored in the controller **110**.

[0099] During actual measurement using the gas sensor **100**, the NO_x concentration identification part **130N** acquires a value of the NO_x detection current **Ip1** from the first measurement pump cell control part **122a**. A value of the concentration of NO_x corresponding to the acquired NO_x detection current **Ip1** is identified with reference to the **Ip1**-NO_x data. The concentration of NO_x in the measurement gas is thereby identified.

[0100] The measurement gas in which NO_x has been reduced in the second chamber **40** to the extent that H₂O and CO₂ are not reduced is introduced into the third chamber **61**. H₂O and CO₂ contained in the measurement gas are reduced in the third chamber **61**. That is to say, oxygen is further pumped out of the measurement gas of which oxygen has been pumped out in the first chamber **20** and in which NO_x has been reduced in the second chamber **40** by operation of the second measurement pump cell **41**, so that a reduction (decomposition) reaction ($2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ and $2\text{CO} \rightarrow 2\text{CO} + \text{O}_2$) of H₂O and CO₂ contained in the measurement gas progresses, and substantially all H₂O and CO₂ are decomposed into hydrogen (H₂), carbon monoxide (CO), and oxygen.

[0101] Reduction (decomposition) of H.sub.2O and CO.sub.2 and pumping-out of oxygen as generated are performed in the way that the second measurement pump cell control part **122b** of the controller **110** sets a target value (control voltage) of the electromotive force **V2** in the third chamber sensor cell **82** to a value in a range of 1000 mV to 1500 mV (preferably 1000 mV) and feedback-controls the voltage **Vp2** applied from the variable power supply **46** to the second measurement pump cell **41** in accordance with a difference between an actual value and the target value of the electromotive force **V2** so that the electromotive force **V2** is maintained at the target value. It is also suggested from the graph of FIG. **4** that the target value of the electromotive force **V2** be preferably set to the value in the range of 1000 mV to 1500 mV.

[0102] The second measurement pump cell **41** operates in this manner, so that the oxygen partial pressure in the third chamber **61** is maintained at a much lower value than the oxygen partial pressure in each of the first chamber **20** and the second chamber **40**. It is approximately 10.sup.-20 atm when an equation $V2=1000\text{ mV}$ holds true, for example. The measurement gas thus no longer substantially contains NO_x, H.sub.2O, CO.sub.2, and oxygen while containing H.sub.2 and CO (and further N.sub.2).

[0103] The measurement gas containing H.sub.2 and CO while not substantially containing NO_x, H.sub.2O, CO.sub.2, and oxygen is introduced into the fourth chamber **63**.

[0104] In the fourth chamber **63**, oxygen is pumped in by operation of the third measurement pump cell **66**, and only H.sub.2 contained in the introduced measurement gas is selectively oxidized.

[0105] Pumping-in of oxygen is performed in the way that the third measurement pump cell control part **122c** of the controller **110** sets a target value (control voltage) of the electromotive force **V3** in the fourth chamber sensor cell **83** to a value in a range of 250 mV to 450 mV (preferably 350 mV) and feedback-controls the voltage **Vp3** applied from the variable power supply **68** to the third measurement pump cell **66** in accordance with a difference between an actual value and the target value of the electromotive force **V3** so that the electromotive force **V3** is maintained at the target value.

[0106] The third measurement pump cell **66** operates in this manner, so that an oxidation (a combustion) reaction $2\text{H.sub.2} + \text{O.sub.2} \rightarrow 2\text{H.sub.2O}$ is facilitated, and H.sub.2O in an amount correlating with the amount of H.sub.2O introduced through the gas inlet **10** is generated again in the fourth chamber **63**. In the present embodiment, H.sub.2O in the correlating amount means that the amount of H.sub.2O introduced through the gas inlet **10** and the amount of H.sub.2O generated again by oxidation of H.sub.2 generated by decomposition of H.sub.2O are the same or are within a certain error range allowable in terms of measurement accuracy.

[0107] The target value of the electromotive force **V3** is set to the value in the range of 250 mV to 450 mV, so that the oxygen partial pressure in the fourth chamber **63** is maintained at a value in a range in which almost all H.sub.2 is oxidized but CO is not oxidized. It is approximately 10.sup.-7 atm when an equation $V3=350\text{ mV}$ holds true, for example.

[0108] Providing the third measurement electrode **64** as the cermet electrode containing the Pt—Au alloy having an Au concentration of 1 wt % or more and 50 wt % or less as the metal component as described above also contributes to improvement in selective H.sub.2 oxidation property.

[0109] In addition, any measures to devise a shape (a width and a thickness), placement (a density), and the like of the heater **72** may be taken to further suppress a rise in temperature of the third measurement electrode **64**.

[0110] In the gas sensor **100** according to the present embodiment operating in the above-mentioned manner, the concentrations of H.sub.2O and CO.sub.2 in the measurement gas are identified based on the oxygen pump current **Ip2** flowing through the second measurement pump cell **41** during pumping-out of oxygen including reduction of H.sub.2O and CO.sub.2 and the oxygen pump current **Ip3** flowing through the third measurement pump cell **66** during pumping-in of oxygen for oxidation of H.sub.2.

[0111] FIGS. **5** and **6** are diagrams showing, when only one of H.sub.2O and CO.sub.2 alone is

contained in the measurement gas as a main sensing target gas component and when H.sub.2O and CO.sub.2 having equal concentrations are contained in the measurement gas as main sensing target gas components, dependence of the oxygen pump current Ip2 and the oxygen pump current Ip3 on a concentration of the sensing target gas component and the concentrations of the sensing target gas components.

[0112] In FIGS. 5 and 6, a graph when H.sub.2O alone is contained as the sensing target gas component is shown by circles, a graph when CO.sub.2 alone is contained as the sensing target gas component is shown by triangles, and a graph when H.sub.2O and CO.sub.2 having equal concentrations are contained as the sensing target gas components (described as “H.sub.2O+CO.sub.2” in the figure) is shown by squares. These graphs are obtained by operating the gas sensor **100** in an atmosphere of each of model gases containing a sensing target gas component having a known concentration and containing oxygen and nitrogen as the balance. The element driving temperature is 800° C. or more, and a temperature of each of the model gases is 200° C.

[0113] As can be seen from FIG. 5, the graph monotonically increases and is approximately linear when only H.sub.2O is contained as the sensing target gas component and when only CO.sub.2 is contained as the sensing target gas component.

[0114] Furthermore, a value of the oxygen pump current Ip2 when H.sub.2O and CO.sub.2 having equal concentrations are contained as the sensing target gas components is the sum of a value of the oxygen pump current Ip2 when H.sub.2O alone is contained and a value of the oxygen pump current Ip2 when CO.sub.2 alone is contained. It has also been confirmed that a value of the oxygen pump current Ip2 when H.sub.2O and CO.sub.2 have different proportions is the sum of a value of the oxygen pump current Ip2 when H.sub.2O alone having a concentration in accordance with its proportion is contained and a value of the oxygen pump current Ip2 when CO.sub.2 alone having a concentration in accordance with its proportion is contained, although it is not shown.

[0115] On the other hand, as shown in FIG. 6, a graph of the oxygen pump current Ip3 when only H.sub.2O is contained as a sensing target gas component monotonically decreases (the absolute value thereof monotonically increases) and is approximately linear. The oxygen pump current Ip3 has a negative value because the oxygen pump current Ip3 flows in a direction in which oxygen is pumped in to oxidize H.sub.2 generated by reduction in the third chamber **61** again, while the direction of the oxygen pump current when the third measurement pump cell **66** pumps out oxygen is assumed to be the positive direction of the oxygen pump current as described above.

[0116] In contrast, a value of the oxygen pump current Ip3 when only CO.sub.2 is contained as a sensing target gas component is maintained approximately zero. This indicates that CO generated by reduction in the third chamber **61** is not oxidized again by operation of the third measurement pump cell **66**.

[0117] A graph of the oxygen pump current Ip3 when H.sub.2O and CO.sub.2 having equal concentrations are contained as the sensing target gas components substantially coincides with the graph of the oxygen pump current Ip3 when H.sub.2O alone is contained. This is consistent with the oxygen pump current Ip3 of approximately zero when only CO.sub.2 is contained as the sensing target gas component. It has also been confirmed that a value of the oxygen pump current Ip3 when H.sub.2O and CO.sub.2 have different proportions substantially coincides with the graph of the oxygen pump current Ip3 when H.sub.2O alone is contained and the graph of the oxygen pump current Ip3 when CO.sub.2 alone is contained, although it is not shown. This means that the oxygen pump current Ip3 virtually depends only on the concentration of H.sub.2O, and thus the concentration of H.sub.2O can be identified once the oxygen pump current Ip3 is known.

[0118] In the present embodiment, the concentrations of H.sub.2O and CO.sub.2 in the measurement gas are measured using properties of the oxygen pump current Ip2 and the oxygen pump current Ip3 as described above. The oxygen pump current Ip2 and the oxygen pump current Ip3 during actual measurement using the gas sensor **100** are hereinafter also referred to as a total

reducing current I_{p2} and a water vapor equivalent current I_{p3} .

[0119] Specifically, prior to use of the gas sensor **100**, characteristics data showing a relationship between the oxygen pump current I_{p2} and a concentration of each gas when the measurement gas contains only one of H_2O and CO_2 and does not contain the other one of H_2O and CO_2 as shown in FIG. 5 (hereinafter also referred to as I_{p2} - H_2O data and I_{p2} - CO_2 data) and characteristics data showing a relationship between the oxygen pump current I_{p3} and a concentration of H_2O when the measurement gas contains H_2O and does not contain CO_2 as shown in FIG. 6 (hereinafter also referred to as I_{p3} - H_2O data) are acquired in advance using model gases having known concentrations, and these pieces of characteristics data are stored in the controller **110**. The I_{p2} - H_2O data and the I_{p2} - CO_2 data respectively have a value indicating a contribution of H_2O and a value indicating a contribution of CO_2 in the total reducing current I_{p2} .

[0120] The oxygen pump current I_{p2} has a value in accordance with diffusion resistance provided to the measurement gas from the gas inlet **10** to the third chamber **61** of the sensor element **101**, and the oxygen pump current I_{p3} has a value in accordance with diffusion resistance provided to the measurement gas from the gas inlet **10** to the fourth chamber **63** of the sensor element **101**. The I_{p2} - H_2O data, the I_{p2} - CO_2 data, and the I_{p3} - H_2O data thus strictly vary with each sensor element **101** of the gas sensor **100**. These pieces of characteristics data are thus preferably identified for each gas sensor **100**. As for gas sensors **100** manufactured under the same condition and from the same lot, however, characteristics data acquired for one particular gas sensor **100** may be applied to another gas sensor **100** from the same lot when it is confirmed that an error is within tolerance.

[0121] When actual measurement is performed using the gas sensor **100**, the measurement gas is introduced into the sensor element **101** heated to the element driving temperature, and the adjustment pump cell **21**, the first measurement pump cell **50**, the second measurement pump cell **41**, and the third measurement pump cell **66** operate in the above-mentioned manner. The water vapor concentration identification part **130H** acquires the water vapor equivalent current I_{p3} from the third measurement pump cell control part **122c** and identifies a concentration of H_2O corresponding to the acquired value based on the I_{p3} - H_2O data.

[0122] Once the concentration of H_2O is identified, the carbon dioxide concentration identification part **130C** acquires a value of the total reducing current I_{p2} from the second measurement pump cell control part **122b** and identifies a contribution of H_2O having the identified concentration in the total reducing current I_{p2} , that is, the amount of current due to reduction of H_2O in the total reducing current I_{p2} based on the I_{p2} - H_2O data. The acquired value is subtracted from the value of the total reducing current I_{p2} to identify a contribution of CO_2 in the total reducing current I_{p2} . A concentration of CO_2 corresponding to the contribution of CO_2 is finally identified based on the I_{p2} - CO_2 data.

[0123] In the gas sensor **100** according to the present embodiment, the concentrations of H_2O and CO_2 in the measurement gas are measured as described above.

[0124] Alternatively, a relationship between the water vapor equivalent current I_{p3} and the oxygen pump current I_{p2} corresponding to the contribution of H_2O in the total reducing current I_{p2} may be identified in advance, characteristics data (hereinafter referred to as H_2O characteristics data) indicating the relationship may be stored in the controller **110**, and the carbon dioxide concentration identification part **130C** may identify the contribution of H_2O in the total reducing current I_{p2} directly from the water vapor equivalent current I_{p3} using the H_2O characteristics data.

[0125] FIG. 7 is a diagram showing an example of the H_2O characteristics data. In FIG. 7, the absolute value of the water vapor equivalent current I_{p3} is shown on an x-axis, and a value of the oxygen pump current I_{p2} corresponding to the contribution of H_2O in the total reducing current I_{p2} is shown on a y-axis. As shown in FIG. 7, there is a linear relationship between the

water vapor equivalent current I_{p3} and the contribution of H_{2O} in the total reducing current I_{p2} , so that a relation expressing the linear relationship is only required to be identified as the H_{2O} characteristics data.

[0126] Alternatively, a value of the y intercept of the relation should theoretically be zero and, in a case of the gas sensor **100** operating normally, is a value small enough to actually be considered zero. Thus, only the slope of the relation expressing the above-mentioned linear relationship may be stored in the controller **110** as the H_{2O} characteristics data, and the carbon dioxide concentration identification part **130C** may use the product of a value of the slope and the water vapor equivalent current I_{p3} as the contribution of H_{2O} in the total reducing current I_{p2} .

[0127] The slope in the H_{2O} characteristics data corresponds to a ratio of the diffusion resistance provided to the measurement gas from the gas inlet **10** to the fourth chamber **63** to the diffusion resistance provided to the measurement gas from the gas inlet **10** to the third chamber **61**.

[0128] The concentration of oxygen is identified using the oxygen pump current I_{p0} flowing through the adjustment pump cell **21** in parallel with identification of the concentrations of NO_x , H_{2O} , and CO_2 .

[0129] In the gas sensor **100** according to the present embodiment, oxygen is pumped out of the measurement gas introduced through the gas inlet **10** in the first chamber **20** by operation of the adjustment pump cell **21** as described above. Oxygen is pumped out so that NO_x , H_{2O} , and CO_2 are not reduced, and the oxygen pump current I_{p0} (hereinafter also referred to as an oxygen detection current I_{p0}) flowing in this case is substantially proportional to the concentration of oxygen contained in the measurement gas introduced through the gas inlet **10**. That is to say, there is a linear relationship between the oxygen detection current I_{p0} and the concentration of oxygen in the measurement gas. Data (I_{p0} - O_2 data) indicating the linear relationship is identified in advance using model gases having known oxygen concentrations and is stored in the controller **110**.

[0130] During actual measurement using the gas sensor **100**, the oxygen concentration identification part **130A** acquires a value of the oxygen detection current I_{p0} from the adjustment pump cell control part **121**. A value of the concentration of oxygen corresponding to the acquired oxygen detection current I_{p0} is identified with reference to the I_{p0} - O_2 data. The concentration of oxygen in the measurement gas is thereby identified.

[0131] As described above, in the gas sensor according to the present embodiment, the concentrations of H_{2O} and CO_2 can be measured when the measurement gas contains H_{2O} and CO_2 as in a conventional gas sensor. In addition, the concentration of NO_x can simultaneously be measured. Furthermore, the concentration of oxygen can also accurately be obtained.

[0132] In addition, in a case of the gas sensor according to the present embodiment, H_{2O} and CO_2 are not reduced in the first chamber to be at a highest temperature during operation in contrast to the gas sensor in conventional technology, so that a voltage applied to the adjustment pump cell pumping out oxygen from the first chamber is suppressed to be lower than that in the gas sensor in conventional technology, thereby to suitably suppress cracking and blackening of the sensor element.

[0133] That is to say, according to the present embodiment, a multi-gas sensor having more long-term reliability than before and capable of simultaneously measuring many types of gases is implemented.

Claims

1. A gas sensor capable of measuring concentrations of a plurality of sensing target gas components, the gas sensor comprising: a sensor element having a structure formed of an oxygen-ion conductive solid electrolyte; and a controller controlling operation of the gas sensor, wherein

the sensor element comprises: a gas inlet through which a measurement gas is introduced; a first chamber, a second chamber, a third chamber, and a fourth chamber communicating sequentially from the gas inlet via different diffusion control parts; an adjustment pump cell including an inner electrode formed to face the first chamber, an out-of-space pump electrode provided at a location other than a location in the first chamber, the second chamber, the third chamber, and the fourth chamber, and a portion of the solid electrolyte present between the inner electrode and the out-of-space pump electrode; a first measurement pump cell including a first measurement electrode formed to face the second chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the first measurement electrode and the out-of-space pump electrode; a second measurement pump cell including a second measurement electrode formed to face the third chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the second measurement electrode and the out-of-space pump electrode; a third measurement pump cell including a third measurement electrode formed to face the fourth chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the third measurement electrode and the out-of-space pump electrode; and a heater heating the sensor element, the inner electrode is a cermet electrode containing a Pt—Au alloy as a metal component, the Pt—Au alloy having an Au concentration of 0.5 wt % or more, the first measurement electrode is another cermet electrode containing a Pt—Rh alloy as a metal component, the adjustment pump cell pumps oxygen out of the measurement gas introduced through the gas inlet into the first chamber so that NO_x, water vapor, and carbon dioxide contained in the measurement gas are not decomposed, the first measurement pump cell pumps out oxygen from the second chamber so that substantially all NO_x contained in the measurement gas introduced from the first chamber into the second chamber is reduced, the second measurement pump cell pumps out oxygen from the third chamber so that substantially all water vapor and carbon dioxide contained in the measurement gas introduced from the second chamber into the third chamber are reduced, the third measurement pump cell pumps oxygen into the fourth chamber to selectively oxidize, in the fourth chamber, hydrogen having been generated by reduction of water vapor and contained in the measurement gas introduced from the third chamber into the fourth chamber, and the controller identifies: a concentration of NO_x contained in the measurement gas based on a magnitude of a NO_x detection current as an oxygen pump current flowing between the first measurement electrode and the out-of-space pump electrode when oxygen is pumped out from the second chamber by the first measurement pump cell; a concentration of water vapor contained in the measurement gas based on a value of a water vapor equivalent current as an oxygen pump current flowing between the third measurement electrode and the out-of-space pump electrode when hydrogen is oxidized with oxygen pumped into the forth chamber by the third measurement pump cell; and a concentration of carbon dioxide contained in the measurement gas based on the value of the water vapor equivalent current and a value of a total reducing current as an oxygen pump current flowing between the second measurement electrode and the out-of-space pump electrode when water vapor and carbon dioxide are reduced by the second measurement pump cell pumping out oxygen from the third chamber.

2. The gas sensor according to claim 1, wherein the controller stores Ip1-NO_x data indicating a relationship between the NO_x detection current and a concentration of NO_x, the Ip1-NO_x data being identified in advance, and identifies the concentration of NO_x contained in the measurement gas based on the NO_x detection current when NO_x contained in the measurement gas is reduced and the Ip1-NO_x data.

3. The gas sensor according to claim 2, wherein the controller further stores: Ip2-H.sub.2O data indicating a relationship between an oxygen pump current flowing through the second measurement pump cell and a concentration of water vapor when the measurement gas contains water vapor and does not contain carbon dioxide, the Ip2-H.sub.2O data being identified in advance; Ip2-CO.sub.2 data indicating a relationship between an oxygen pump current flowing through the second measurement pump cell and a concentration of carbon dioxide when the

measurement gas contains carbon dioxide and does not contain water vapor, the Ip2-CO.sub.2 data being identified in advance; and Ip3-H.sub.2O data indicating a relationship between an oxygen pump current flowing through the third measurement pump cell and a concentration of water vapor when the measurement gas contains water vapor and does not contain carbon dioxide, the Ip3-H.sub.2O data being identified in advance, and identifies: a concentration of water vapor corresponding to the value of the water vapor equivalent current in the Ip3-H.sub.2O data as the concentration of water vapor contained in the measurement gas, and a contribution of water vapor by reduction in the total reducing current based on the identified concentration of water vapor contained in the measurement gas and the Ip2-H.sub.2O data, and a concentration of carbon dioxide corresponding to a difference value obtained by subtracting the contribution from the total reducing current in the Ip2-CO.sub.2 data as the concentration of carbon dioxide contained in the measurement gas.

4. The gas sensor according to claim 2, wherein the controller further stores: Ip2-CO.sub.2 data indicating a relationship between an oxygen pump current flowing through the second measurement pump cell and a concentration of carbon dioxide when the measurement gas contains carbon dioxide and does not contain water vapor, the Ip2-CO.sub.2 data being identified in advance; Ip3-H.sub.2O data indicating a relationship between an oxygen pump current flowing through the third measurement pump cell and a concentration of water vapor when the measurement gas contains water vapor and does not contain carbon dioxide, the Ip3-H.sub.2O data being identified in advance; and H.sub.2O characteristics data indicating a relationship between the water vapor equivalent current and an oxygen pump current corresponding to a contribution of water vapor in the total reducing current, the H.sub.2O characteristics data being identified in advance, and identifies: a concentration of water vapor corresponding to the value of the water vapor equivalent current in the Ip3-H.sub.2O data as the concentration of water vapor contained in the measurement gas, a contribution of water vapor by reduction in the total reducing current based on the water vapor equivalent current and the H.sub.2O characteristics data, and a concentration of carbon dioxide corresponding to a difference value obtained by subtracting the contribution from the total reducing current in the Ip2-CO.sub.2 data as the concentration of carbon dioxide contained in the measurement gas.

5. The gas sensor according to claim 1, wherein the controller further identifies a concentration of oxygen contained in the measurement gas based on a magnitude of a current flowing between the inner electrode and the out-of-space pump electrode when the adjustment pump cell pumps out oxygen from the first chamber.

6. The gas sensor according to of claim 1, wherein the third measurement electrode is a cermet electrode containing a Pt—Au alloy as a metal component, and the Pt—Au alloy has an Au concentration of 1 wt % or more and 50 wt % or less.

7. A concentration measurement method of measuring concentrations of a plurality of sensing target gas components using a gas sensor, wherein the gas sensor includes a sensor element having an elongated planar structure formed of an oxygen-ion conductive solid electrolyte, the sensor element comprises: a gas inlet through which a measurement gas is introduced; a first chamber, a second chamber, a third chamber, and a fourth chamber communicating sequentially from the gas inlet via different diffusion control parts; an adjustment pump cell including an inner electrode formed to face the first chamber, an out-of-space pump electrode provided at a location other than a location in the first chamber, the second chamber, the third chamber, and the fourth chamber, and a portion of the solid electrolyte present between the inner electrode and the out-of-space pump electrode; a first measurement pump cell including a first measurement electrode formed to face the second chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the first measurement electrode and the out-of-space pump electrode; a second measurement pump cell including a second measurement electrode formed to face the third chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between

the second measurement electrode and the out-of-space pump electrode; a third measurement pump cell including a third measurement electrode formed to face the fourth chamber, the out-of-space pump electrode, and a portion of the solid electrolyte present between the third measurement electrode and the out-of-space pump electrode; and a heater heating the sensor element, the inner electrode is a cermet electrode containing a Pt—Au alloy as a metal component, the Pt—Au alloy having an Au concentration of 0.5 wt % or more, the first measurement electrode is another cermet electrode containing a Pt—Rh alloy as a metal component, and the concentration measurement method using the gas sensor comprises: a) pumping, using the adjustment pump cell, oxygen out of the measurement gas introduced through the gas inlet into the first chamber so that NO_x, water vapor, and carbon dioxide contained in the measurement gas are not decomposed; b) pumping out, using the first measurement pump cell, oxygen from the second chamber so that substantially all NO_x contained in the measurement gas introduced from the first chamber into the second chamber is reduced; c) pumping out, using the second measurement pump cell, oxygen from the third chamber so that substantially all water vapor and carbon dioxide contained in the measurement gas introduced from the second chamber into the third chamber are reduced; d) pumping, using the third measurement pump cell, oxygen into the fourth chamber to selectively oxidize, in the fourth chamber, hydrogen having been generated by reduction of water vapor and contained in the measurement gas introduced from the third chamber into the fourth chamber; e) identifying a concentration of NO_x contained in the measurement gas based on a magnitude of a NO_x detection current as an oxygen pump current flowing between the first measurement electrode and the out-of-space pump electrode when NO_x is reduced by pumping out oxygen from the second chamber using the first measurement pump cell; f) identifying a concentration of water vapor contained in the measurement gas based on a value of a water vapor equivalent current as an oxygen pump current flowing between the third measurement electrode and the out-of-space pump electrode when hydrogen is oxidized with oxygen pumped into the fourth chamber using the third measurement pump cell; and g) identifying a concentration of carbon dioxide contained in the measurement gas based on the value of the water vapor equivalent current and a value of a total reducing current as an oxygen pump current flowing between the second measurement electrode and the out-of-space pump electrode when water vapor and carbon dioxide are reduced by pumping out oxygen from the third chamber using the second measurement pump cell.

8. The concentration measurement method using the gas sensor according to claim 7, further comprising h) prior to the steps a) to g), identifying Ip1-NO_x data in advance, the Ip1-NO_x data indicating a relationship between the NO_x detection current and a concentration of NO_x, wherein in the step e), the concentration of NO_x contained in the measurement gas is identified based on the NO_x detection current when NO_x contained in the measurement gas is reduced and the Ip1-NO_x data.

9. The concentration measurement method using the gas sensor according to claim 8, wherein the step h) further comprises identifying: Ip2-H.sub.2O data indicating a relationship between an oxygen pump current flowing through the second measurement pump cell and a concentration of water vapor when the measurement gas contains water vapor and does not contain carbon dioxide; Ip2-CO.sub.2 data indicating a relationship between an oxygen pump current flowing through the second measurement pump cell and a concentration of carbon dioxide when the measurement gas contains carbon dioxide and does not contain water vapor; and Ip3-H.sub.2O data indicating a relationship between an oxygen pump current flowing through the third measurement pump cell and a concentration of water vapor when the measurement gas contains water vapor and does not contain carbon dioxide, in the step f), a concentration of water vapor corresponding to the value of the water vapor equivalent current in the Ip3-H.sub.2O data is identified as the concentration of water vapor contained in the measurement gas, and in the step g), a contribution of water vapor by reduction in the total reducing current is identified based on the concentration of water vapor contained in the measurement gas identified in the step f) and the Ip2-H.sub.2O data, and a

concentration of carbon dioxide corresponding to a difference value obtained by subtracting the contribution from the total reducing current in the Ip2-CO.sub.2 data is identified as the concentration of carbon dioxide contained in the measurement gas.

10. The concentration measurement method using the gas sensor according to claim 8, wherein the step h) further comprises identifying: Ip2-CO.sub.2 data indicating a relationship between an oxygen pump current flowing through the second measurement pump cell and a concentration of carbon dioxide when the measurement gas contains carbon dioxide and does not contain water vapor; Ip3-H.sub.2O data indicating a relationship between an oxygen pump current flowing through the third measurement pump cell and a concentration of water vapor when the measurement gas contains water vapor and does not contain carbon dioxide; and H.sub.2O characteristics data indicating a relationship between the water vapor equivalent current and an oxygen pump current corresponding to a contribution of water vapor in the total reducing current, in the step f), a concentration of water vapor corresponding to the value of the water vapor equivalent current in the Ip3-H.sub.2O data is identified as the concentration of water vapor contained in the measurement gas, and in the step g), a contribution of water vapor by reduction in the total reducing current is identified based on the water vapor equivalent current and the H.sub.2O characteristics data, and a concentration of carbon dioxide corresponding to a difference value obtained by subtracting the contribution from the total reducing current in the Ip2-CO.sub.2 data is identified as the concentration of carbon dioxide contained in the measurement gas.

11. The concentration measurement method using the gas sensor according to claim 7, further comprising i) identifying a concentration of oxygen contained in the measurement gas based on a magnitude of a current flowing between the inner electrode and the out-of-space pump electrode when oxygen is pumped out from the first chamber using the adjustment pump cell.

12. The concentration measurement method using the gas sensor according to claim 7, wherein the third measurement electrode is a cermet electrode containing a Pt—Au alloy as a metal component, and the Pt—Au alloy has an Au concentration of 1 wt % or more and 50 wt % or less.

13. The gas sensor according to claim 2, wherein the controller further identifies a concentration of oxygen contained in the measurement gas based on a magnitude of a current flowing between the inner electrode and the out-of-space pump electrode when the adjustment pump cell pumps out oxygen from the first chamber.

14. The gas sensor according to claim 3, wherein the controller further identifies a concentration of oxygen contained in the measurement gas based on a magnitude of a current flowing between the inner electrode and the out-of-space pump electrode when the adjustment pump cell pumps out oxygen from the first chamber.

15. The gas sensor according to claim 4, wherein the controller further identifies a concentration of oxygen contained in the measurement gas based on a magnitude of a current flowing between the inner electrode and the out-of-space pump electrode when the adjustment pump cell pumps out oxygen from the first chamber.

16. The gas sensor according to of claim 2, wherein the third measurement electrode is a cermet electrode containing a Pt—Au alloy as a metal component, and the Pt—Au alloy has an Au concentration of 1 wt % or more and 50 wt % or less.

17. The gas sensor according to of claim 3, wherein the third measurement electrode is a cermet electrode containing a Pt—Au alloy as a metal component, and the Pt—Au alloy has an Au concentration of 1 wt % or more and 50 wt % or less.

18. The gas sensor according to of claim 4, wherein the third measurement electrode is a cermet electrode containing a Pt—Au alloy as a metal component, and the Pt—Au alloy has an Au concentration of 1 wt % or more and 50 wt % or less.

19. The concentration measurement method using the gas sensor according to claim 8, further comprising i) identifying a concentration of oxygen contained in the measurement gas based on a magnitude of a current flowing between the inner electrode and the out-of-space pump electrode

when oxygen is pumped out from the first chamber using the adjustment pump cell.

20. The concentration measurement method using the gas sensor according to claim 9, further comprising i) identifying a concentration of oxygen contained in the measurement gas based on a magnitude of a current flowing between the inner electrode and the out-of-space pump electrode when oxygen is pumped out from the first chamber using the adjustment pump cell.
